

The Transcription Factor Myc Controls Metabolic Reprogramming upon T Lymphocyte Activation

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SUMMARY

To fulfill the bioenergetic and biosynthetic demand of proliferation, T cells reprogram their metabolic pathways from fatty acid β -oxidation and pyruvate oxidation via the TCA cycle to the glycolytic, pentose-phosphate, and glutaminolytic pathways. Two of the top-ranked candidate transcription factors potentially responsible for the activation-induced T cell metabolic transcriptome, HIF1 α and Myc, were induced upon T cell activation, but only the acute deletion of Myc markedly inhibited activation-induced glycolysis and glutaminolysis in T cells. Glutamine deprivation compromised activation-induced T cell growth and proliferation, and this was partially replaced by nucleotides and polyamines, implicating glutamine as an important source for biosynthetic precursors in active T cells. Metabolic tracer analysis revealed a Myc-dependent metabolic pathway linking glutaminolysis to the biosynthesis of polyamines. Therefore, a Myc-dependent global metabolic transcriptome drives metabolic reprogramming in activated, primary T lymphocytes. This may represent a general mechanism for metabolic reprogramming under patho-physiological conditions.

INTRODUCTION

Proliferating animal cells consume considerable amounts of energy and require de novo synthesis of macromolecules to support their growth and proliferation (DeBerardinis et al., 2008; Newsholme et al., 1985b). Consequently, one of the fundamental problems in animal cells is coordination of the metabolic program with cell growth- and proliferation-associated bioenergetic demand (Jorgensen and Tyers, 2004). Whereas prokaryotes and unicellular eukaryotes rely primarily on homeostatic regulation of cell metabolism, metabolic programs in multicellular eukaryotes have evolved to be actively controlled by extracellular signals (DeBerardinis et al., 2008).

Upon stimulation of antigen receptors, quiescent naive T cells undergo a growth phase of ~24 hr, followed by massive clonal

expansion and differentiation phases that are essential for appropriate immune defense and regulation. During proliferation, T cells can undergo a cell division every 4–6 hr (van Stipdonk et al., 2003). Coinciding with the clearance of antigen, effector T cells undergo an abrupt contraction phase in which the majority of T cells are eliminated by apoptosis (Green, 2003; Schumacher et al., 2010). Accumulation of cell biomass during the initial growth and rapid proliferation during the expansion phase is associated with dramatically increased bioenergetic and biosynthetic demand. Thus, T cell activation provides a unique physiological cellular model to help us understand how animal cells coordinate their metabolic program with cell growth and proliferation in response to extracellular fate-decisi-

g signals. Early studies showed that polyclonal mitogens such as concanavalin A and phytohemagglutinin drive the upregulation of glycolysis and glutamine oxidation in thymocytes and mature lymphocytes (Brand et al., 1984; Newsholme et al., 1985b). Consistent with these observations, the uptake of glutamine and glucose and the consumption of the latter, mainly through glycolysis, are substantially upregulated upon stimulation with anti-CD3 and anti-CD28 (Carr et al., 2010; Frauwirth et al., 2002). Signaling via ERK and AKT pathways has been implicated in the uptake of glutamine and glucose, respectively (Carr et al., 2010; Frauwirth et al., 2002; Wofford et al., 2008), and the mammalian target of rapamycin (mTOR), an essential component of PI3K-Akt pathway, has recently been implicated in regulating fatty acid metabolism in memory T cells (Pearce et al., 2009; Araki et al., 2009). However, the robustness of metabolic changes after T cell activation suggests the presence of additional regulatory steps in the T cell metabolic program, and the molecular mechanisms behind these remain to be explored. The metabolic changes associated with T cell activation are reminiscent of the characteristic metabolic activities of tumor cells and may represent a general metabolic reprogramming during cell growth and proliferation (Kim and Dang, 2006; Vander Heiden et al., 2009; Warburg, 1956).

RESULTS

T Cell Activation Induces Metabolic Reprogramming

To understand how T cells reprogram metabolic pathways to fulfill the bioenergetic and biosynthetic demand for the burst of proliferation upon activation, we used a well-established ex vivo

model to recapitulate the process of antigen stimulation of T cells without a requirement for additional cells. Freshly isolated murine primary T cells were either maintained in interleukin 7 (IL-7) (hereafter “resting T cells”) (Vella et al., 1997) or were stimulated with anti-CD3 plus anti-CD28 (hereafter “active T cells”). After activation, both CD4⁺ and CD8⁺ T cells increased their cell size over a 24 hr period (Figure S1A available online) followed by rapid cell division over the ensuing 24–72 hr (Figure S1B). This provided a 24 hr “window” in which to interrogate changes in metabolism upon activation, without the potentially confounding effects of proliferation.

We applied a mass-spectrum-based metabolomic approach to determine the relative intracellular concentration of metabolites in T cells after activation. Although at an early time point (6 hr after activation), activated T cells contained lower amounts of metabolites than resting T cells, T cells that had been activated for longer times (24 and 30 hr) accumulated markedly higher amounts of metabolites involved in the anabolic pathways of lipids, amino acids, and nucleotides (Figure 1A; Table S1). This may reflect cell growth before entering into the cell cycle (Figure S1A). In contrast, carnitines, which are required for fatty acid β -oxidation, were downregulated after activation (Figure 1A).

Consistent with earlier studies (Frauwirth et al., 2002), activation of T cells dramatically upregulated glycolysis (Figure 1B). In contrast, both mitochondria-dependent fatty acid β -oxidation (FAO) and pyruvate oxidation through the tricarboxylic acid cycle (TCA cycle) were markedly downregulated upon activation (Figures 1C and 1D). Similarly, glutamine consumption through oxidative catabolism was significantly upregulated upon activation (Figure 1E), as suggested by earlier observations (Newsholme et al., 1985a). Glucose consumption via the pentose phosphate pathway (PPP) (Figure 1F) and oxygen consumption (Figure S1C) were also significantly elevated in activated T cells. Therefore, activation rapidly switches T cell metabolism from FAO and pyruvate oxidation via the TCA cycle to aerobic glycolysis, PPP, and glutamine oxidation (hereafter “activation-induced metabolic reprogramming”).

Glucose and Glutamine Catabolism Are Required for Activation-Induced T Cell Proliferation In Vitro and In Vivo

To explore the impact of metabolic reprogramming on T cell growth and proliferation, we stimulated T cells in either complete or nutrient-free media. Whereas deprivation of glutamine but not glucose led to an impairment of activation-induced cell growth (Figure S1D) associated with a reduction of lipid and protein biosynthesis (Figures S1E and S1F), neither condition affected viability or appearance of the activation markers CD25 and CD69 (Figures S1G and S1H). However, both conditions resulted in impaired activation-induced cell proliferation (Figure S1I). Consistent with previous work (Rowell and Wells, 2006), T cell activation was associated with downregulation of the cell cycle inhibitor p27, the upregulation of cyclins and cyclin-dependent kinases (CDKs), and the appearance of markers of S phase, such as the phosphorylation of RB and Cdc6 (Figure S1J). The stabilization of p27 and the absence of S phase markers indicated a G1 cell cycle arrest under glutamine starvation. However, the similar expression and phosphorylation pattern of cell cycle regulators in control and glucose-starved cells sug-

gested that glucose deprivation might inhibit cell proliferation through a mechanism distinct from that of glutamine starvation.

We next applied a panel of potent chemical inhibitors targeting different metabolic pathways. Inhibition of FAO by the carnitine palmitoyltransferase-I (CPT-I) inhibitor Etomoxir (Eto) did not affect T lymphocyte proliferation (Figure S1K), consistent with the low FAO flux in active T cells. In contrast, the hexokinase (HK) inhibitor 2-deoxy-D-glucose (2DG), the glucose-6-phosphate dehydrogenase (G6PD) inhibitor dehydroepiandrosterone (DHEA), the glutaminase (Gls) inhibitor 6-diazo-5-oxo-L-norleucine (DON), or the transaminase inhibitor aminooxyacetate (AOA), each of which inhibits activation-induced metabolic pathways, blocked T cell proliferation (Figure S1K).

To further test the dependency of T cell proliferation on glucose and glutamine catabolism in vivo, we applied 2DG and DON in an adoptive transfer model by using ovalbumin (OVA)-specific, TCR-transgenic OT-II T cells (Figure 1G; Shi et al., 2011). In agreement with our in vitro results, both 2DG and DON partially inhibited the proliferation of antigen-specific active T cells after OVA immunization in vivo (Figure 1H). Therefore, both glucose and glutamine catabolic pathways are required for activation-induced T cell proliferation in vitro and in vivo.

T Cell Metabolic Reprogramming Is Associated with Global Changes in the Metabolic Gene Transcriptome

Antigen activation of T lymphocytes drives distinct transcriptional programs including upregulation of cell cycle and cytokine genes (Lu et al., 2004; Teague et al., 1999). Real-time quantitative PCR (qPCR) analysis revealed a time-dependent upregulation of mRNAs encoding metabolic enzymes and transporters involved in glycolysis (Figure 2A) and in both arms of the PPP (Figure S2A). Notably, only one member in each metabolic enzyme gene family was selectively upregulated (Figure 2A). Immunoblot analysis further confirmed the upregulation of selected metabolic genes at the protein level (Figure S2B).

Glutamine catabolism is not only tightly coupled to different biosynthetic pathways but also generates the anaplerotic substrate α -ketoglutarate (α -KG), which can be metabolized through the TCA cycle to generate either citrate or pyruvate (“glutaminolysis”) (Newsholme et al., 1985a). Whereas the conversion of glutamine (Gln) to glutamate (Glu) is controlled by glutamine-fructose-6-phosphate transaminase 1 (Gfpt1), carbamoyl-phosphate synthetase (CAD), phosphoribosyl pyrophosphate amidotransferase (Ppat), and glutaminase 2 (Gls2), the conversion of glutamate to α -KG is controlled by glutamate dehydrogenase 1 (Glu1), glutamate oxaloacetate transaminase (Got), and ornithine aminotransferase (Oat). Consistent with the idea that α -KG is an anaplerotic substrate of the TCA cycle, metabolic enzymes in the TCA cycle including isocitrate dehydrogenase-3 (IDH3), malate dehydrogenase (MDH), succinate dehydrogenase complex subunit C (SDHc), and malic enzyme 2 (ME2) were also upregulated (Figures 2B, S2A, and S2B).

We compared our metabolic gene expression data of active mouse T lymphocytes with metabolic gene expression data from a recently published microarray study of active human T lymphocytes (Wang et al., 2008). A marked correlation between these two data sets, indicated by the Spearman rank-order correlation coefficient, suggests that metabolic genes in activated human T lymphocytes are similarly regulated (Figure S2C).

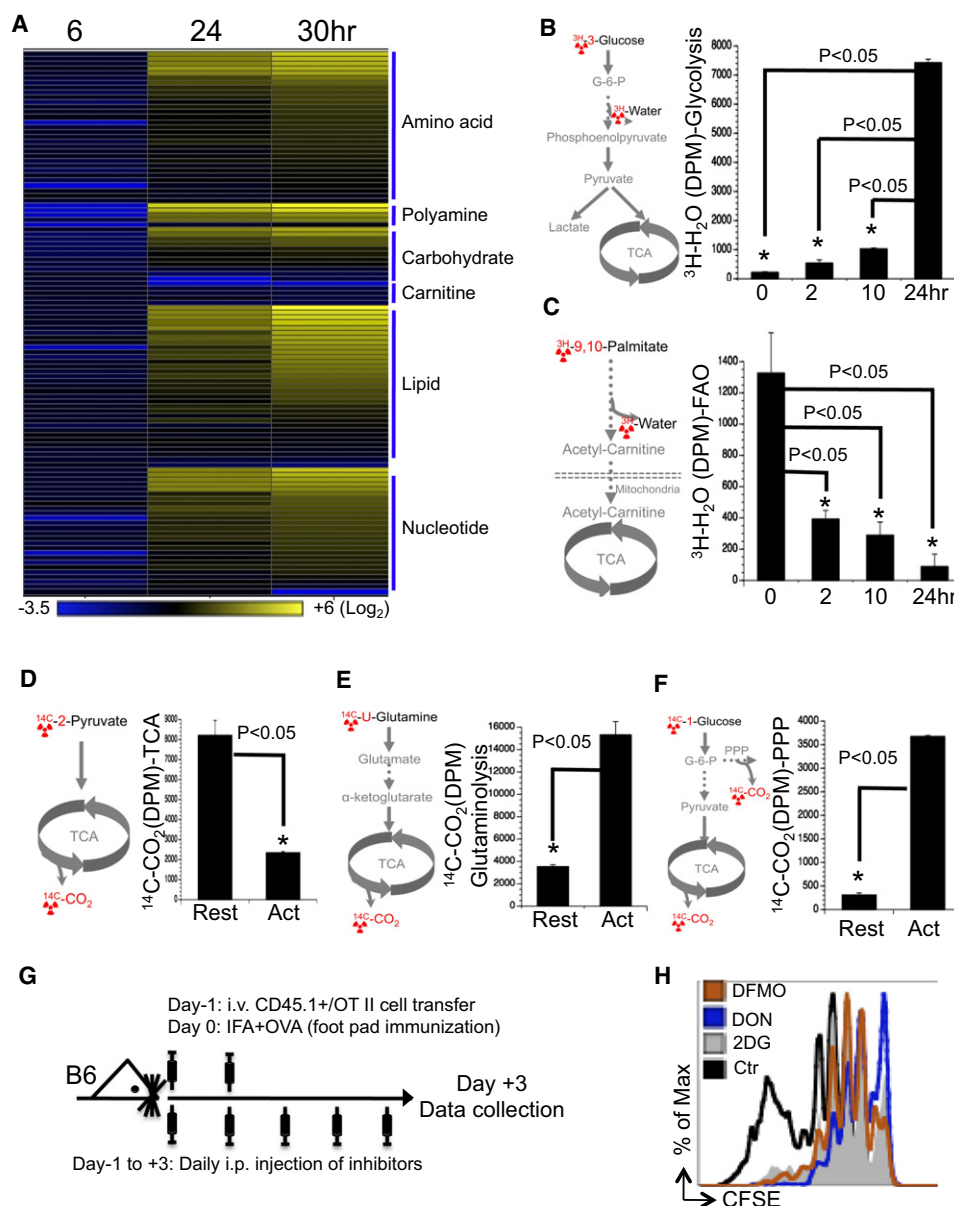


Figure 1. Stimulation of TCR and CD28 Drives T Cell Metabolic Reprogramming

(A) Intracellular metabolites in T cells collected at the indicated times after activation were profiled by mass spectroscopy. The value for each metabolite represents the average of triplicates and the amount of each metabolite in resting T cells was set to 1. The heat map represents the log₂ value of the relative amount of each metabolite, which was grouped in the indicated metabolic pathways (see color scale). The complete metabolomic profile is provided in [Table S1](#). (B–F) Metabolic assays, with the isotopically labeled tracer used highlighted in red (left). Resting T cells (Rest) and activated T cells (collected at 24 hr after activation in D–F or at the indicated times after activation in B and C) were used for measuring the generation of ³H₂O from [3-³H]-glucose (glycolysis, right panel in B) or from [9,10-³H]-palmitic acid (FAO, right panel in C); the generation of ¹⁴CO₂ from [2-¹⁴C]-pyruvate (TCA, right panel in D), from [U-¹⁴C]-glutamine (glutaminolysis, right panel in E), or from [1-¹⁴C]-glucose (PPP, right panel in F). Error bars represent standard deviation from the mean of triplicate samples. Data are representative of three independent experiments.

(G) Overview of the experimental procedure used for assessing OT-II T cell proliferation in vivo (H).

(H) Naive OT-II T cells (CD45.1⁺) were CFSE labeled and transferred into C57BL/6 mice, which were immunized with OVA and treated daily with the indicated inhibitors or vehicle controls. Splenocytes were analyzed 3 days after immunization and OT-II cell proliferation was determined by flow cytometry.

Data represent two independent experiments. See also [Figure S1](#).

Taken together, these results are consistent with our metabolic flux data and indicate that global regulation of metabolic gene transcription is involved in activation-induced T cell metabolic reprogramming.

Myc, but Not HIF1α, Is Required for Activation-Induced T Cell Metabolic Reprogramming

With an *in silico* approach ([Roeder et al., 2009](#)), transcription factors possibly involved in activation-induced T cell metabolic

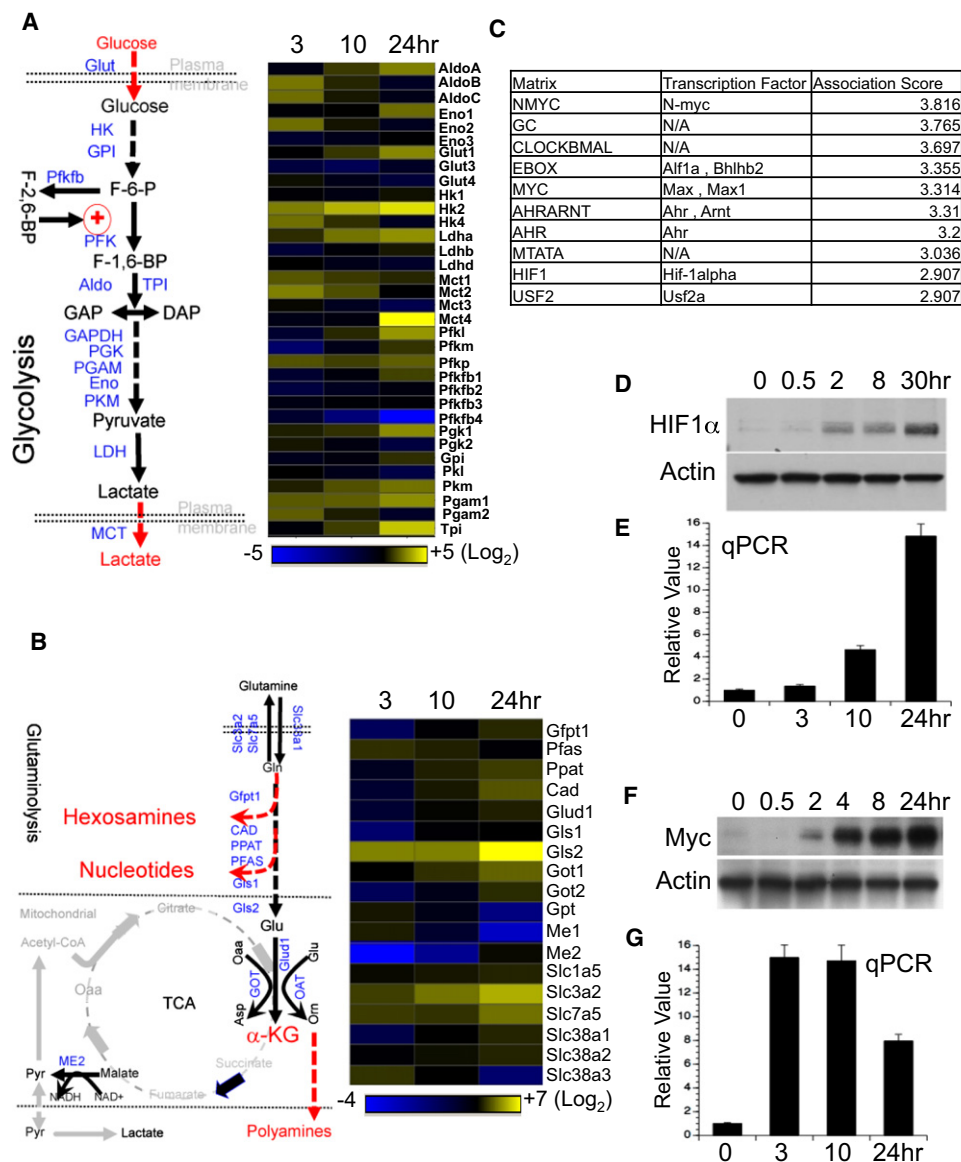


Figure 2. T Cell Activation Drives the Transcription of a Distinct Set of Metabolic Genes

(A and B) The glycolytic pathway (left in A) and the glutaminolytic pathway (left in B), with the metabolic genes measured highlighted in blue. RNA was isolated from T cells collected at the indicated times after activation and used for qPCR analyses of metabolic genes in the glycolytic pathway (right in A) and the glutaminolytic pathway (right in B). mRNA levels in resting T cells were set to 1. The heat map represents the log2 value of the relative mRNA expression level (see color scale). Values and standard deviations are provided in [Table S2](#). Data are representative of two independent experiments.

(C) The top ten predicted corresponding transcription factors and motifs were ranked in descending order of association score, which indicates the likelihood of the binding of a particular transcription factor to the input gene promoters correlating with the expression levels. The input gene list and expression profile is provided in [Table S3](#).

(D–G) Protein and mRNA expression of HIF1 α (D and E, respectively) and of Myc (F and G, respectively) in T cells collected at the indicated times after activation were determined by immunoblot and qPCR. mRNA levels in resting T cells were set to 1. Error bars represent standard deviation from the mean of triplicate samples. Data are representative of two independent experiments.

See also [Figure S2](#).

reprogramming were ranked by the likelihood of a particular transcription factor binding to the input gene promoters (Figure 2C; Table S3). Among the top ten candidates (Figure 2C), hypoxia inducible factor 1, alpha subunit (HIF1 α), and myelocytomatosis oncogene (Myc) were chosen for further investigation. Both HIF1 α and Myc were recently suggested to regulate

cell metabolism in transformed cells (Dang and Semenza, 1999; Gordan et al., 2007) and were highly induced at both the transcription and protein levels upon T cell activation (Figures 2D–2G). The induction of these two transcription factors preceded the induction of most metabolic genes examined (Figures 2A and 2B) and the upregulation of glycolysis and

glutaminolysis (Figures 1B and 1E). In contrast, the unperturbed expression of the endogenous Myc antagonist Mad suggests that this may not be instrumental in the observed changes (Figure S2D).

Early studies have implicated the involvement of ERK-, AKT-, and mTOR-mediated signaling pathways in regulating T cell metabolism (Carr et al., 2010; Frauwirth et al., 2002; Pearce et al., 2009). We next applied a panel of potent chemical inhibitors targeting different activation-induced signaling pathways and followed the expression of Myc and HIF1 α . Inhibition of any of these kinases significantly reduced the induction of Myc and HIF1 α after T cell activation (Figure S2F), which was associated with the reduction of glycolytic activity (Figure S2E). Therefore, ERK-, AKT-, and mTOR-mediated signaling pathways may regulate T cell metabolism at least partially through regulating the expression of Myc and HIF1 α .

We generated a mouse model carrying a conditional HIF1 α allele (*Hif1a^{flox/flox}*) and a tamoxifen-inducible Cre recombinase (CreERT2) transgene (de Luca et al., 2005; Ryan et al., 2000) to delete HIF1 α flox alleles in an acute manner. Freshly isolated T cells were cultured with IL-7 for 2 days with or without 4-Hydroxytamoxifen (4OHT), following which cells were either maintained in IL-7 (resting) or activated with anti-CD3 and anti-CD28 (active) (Figure S3A). As controls, comparable cell proliferation and glycolytic rate were observed in T cells containing the CreER transgene only, and treated with vehicle or 4OHT (Figures S3B and S3C). Efficient deletion of HIF1 α was observed at the protein (Figure 3A) and mRNA (Figure 3B) levels after 4OHT treatment. However, glycolysis, glutaminolysis, and FAO were unaffected by acute HIF1 α deletion 24 hr after T cell activation (Figures 3C, 3D, and S3E). Whereas the induction of two known HIF1 α target genes, LDHA and HK2, was moderately impaired by the deletion of HIF1 α , the expression of other activation-induced metabolic genes was comparable in WT and HIF1 α -deficient T cells (Figure S3F). Moreover, activation-associated cell proliferation (Figure 3E), cell growth (Figure 3F), and cell surface appearance of the activation marker CD25 (Figure S3D) were unaffected in HIF1 α -deficient cells, and mild impairment of glycolysis was observed only at 72 hr postactivation (Figure S3G). Therefore, HIF1 α is not required for activation-induced metabolic reprogramming prior to T cell proliferation, but may be involved in optimally maintaining T cell glycolytic activities after entry into the cell cycle.

We then generated mice carrying a conditional Myc allele (*Myc^{flox/flox}*) and a tamoxifen-inducible Cre recombinase (CreERTam) (Badea et al., 2003; de Alboran et al., 2001). After addition of 4OHT, efficient deletion of the Myc gene was observed at the protein (Figure 4A) and mRNA (Figure 4B) levels. Although Myc was suggested to be required for T cell proliferation and growth under most experimental conditions (Dose et al., 2006; Iritani et al., 2002), one study indicated that it is essential for cell proliferation but not cell growth (Trumpp et al., 2001). We found that acute deletion of Myc blocked activation-associated cell proliferation and growth in vitro (Figures 4C and 4D). Metabolomic profiling further revealed that Myc-deficient T cells accumulated significantly lower levels of lipids, amino acids, and nucleotides (Figure 4E). We then examined the role of Myc in regulating T cell proliferation and growth in vivo in response to staphylococcal enterotoxin B (SEB) (Figure 4F),

which as a superantigen engaged sufficiently large numbers of naive V β 8⁺ T cells to perform metabolic studies (see below). Consistent with our in vitro results, the in vivo deletion of Myc blocked SEB-induced V β 8⁺ T cell proliferation and growth (Figures 4G and 4H). We observed that acute depletion of Myc, in vivo, did not compromise the health of the animals over a period of 1 week, consistent with a recent report (Soucek et al., 2008).

Myc regulates cell cycle progression either directly or indirectly through controlling the expression of several essential cell cycle regulators including p27, cyclins, and CDKs (Dang et al., 2006; Eilers and Eisenman, 2008). Whereas the downregulation of p27 and upregulation of Cyclin D3 were largely undisturbed, induction of Cyclin A, cdk2, cdk4, and cdc25a were impaired to varying degrees in Myc-deficient T cells (Figure S4A). Phosphorylation of RB and Cdc6, two S phase markers, was delayed and abolished, respectively (Figure S4A). In contrast, the upregulation of the activation markers CD69 and CD25 (Figure S4B) was comparable between Myc-deficient and WT cells, as was the phosphorylation of AKT and ERK (Figure S4C). The expression of several cytokines was unaffected or elevated in Myc-deficient cells (Figures S4D and S4E). Thus, the general signaling pathways mediated by TCR and cytokines are largely intact in T cells lacking Myc, and the acute deletion of Myc abrogates activation-induced T cell growth and proliferation but not other activation-induced events in T cells.

Myc-Dependent Induction of Glucose Metabolic Genes Drives Glucose Catabolism

Deletion of Myc significantly impaired activation-induced glycolytic flux at both early and late time points (Figures 5A and S5A). Consistent with this, intracellular lactate in Myc-deficient T cells was lower than in WT control cells (Figure S5B). Moreover, the partial reduction of Myc expression in T cells containing heterozygous *Myc^{flox/wt}* alleles led to a moderate impairment of activation-induced glycolysis (Figures S5C and S5D). This suggests that the maximum expression of Myc is required for the maximum increase in glycolysis after activation of T cells. The upregulation of several glycolytic enzymes and transporters was impaired to varying degrees in Myc-deficient T cells upon activation (Figure 5B). Further, Myc deficiency led to a reduction of HK2 and pyruvate kinase muscle isoform 2 (PKM2) at the protein level (Figure S5F). Finally, we tested the relevance of Myc in regulating T cell metabolism in response to SEB, as described above (Figure 4F). Consistent with our in vitro results, Myc is required for the induction of enhanced glycolytic activity and metabolic gene expression in SEB-responsive T cells in vivo (Figures 5C and 5D).

Because glycolysis interconnects with other metabolic pathways, such as the PPP, the TCA cycle, and FAO, we investigated the metabolic flux through these pathways. Acute deletion of Myc moderately inhibited activation-induced upregulation of metabolic activity through the PPP (Figure 5E), and the induction of two key metabolic enzymes in the PPP, Tkt and G6PDx, was compromised in Myc-deficient T cells upon activation (Figure S5E). In contrast, pyruvate metabolic flux, FAO flux, and oxygen consumption rates were unaffected by Myc deletion (Figures 5F, S5G, and S5H).

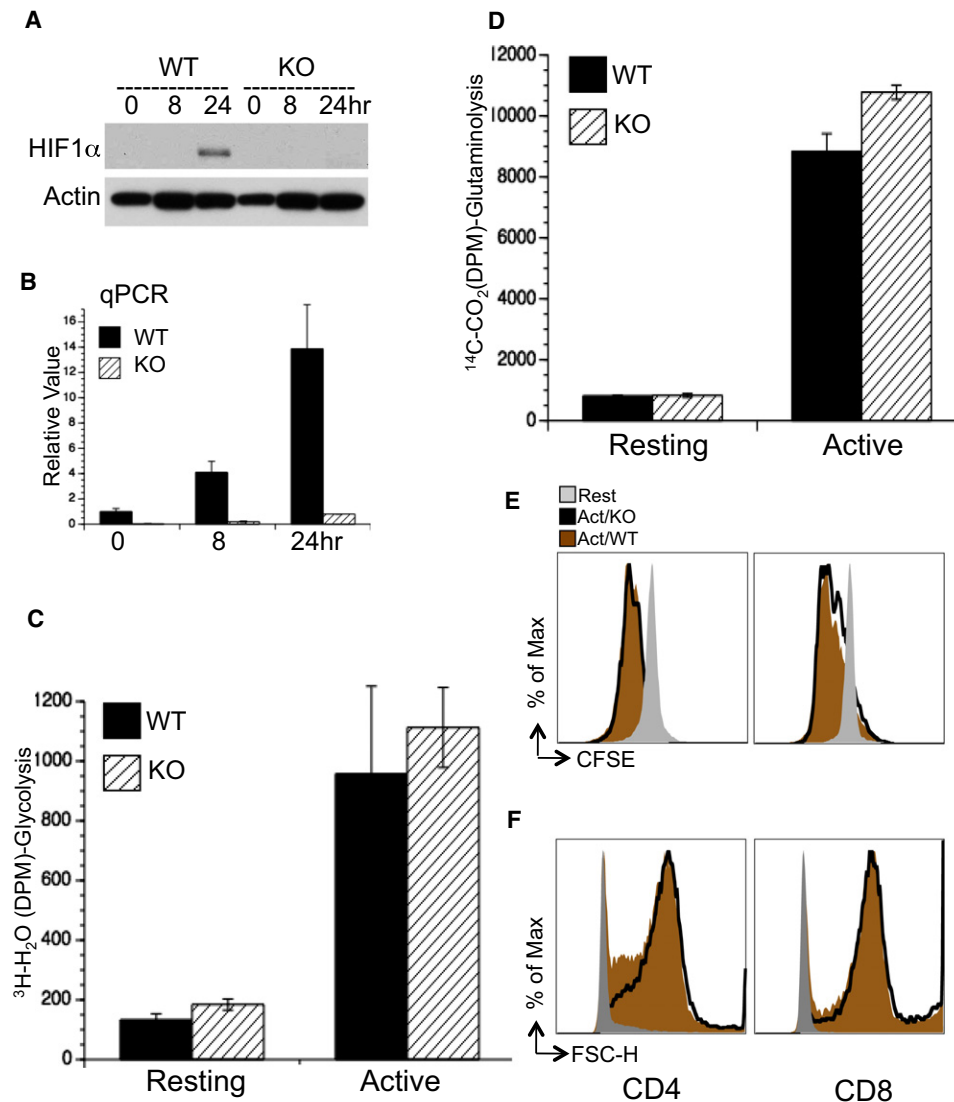


Figure 3. HIF1 α Is Not Required for Activation-Induced Metabolic Reprogramming

(A and B) T cells were isolated from RosaCreERT2/HIF1 $\alpha^{lox/lox}$ mice and pretreated with vehicle (WT) or with 500 nM 4OHT (KO). HIF1 α protein and mRNA levels in T cells collected at the indicated times after activation were determined by immunoblot (A) and qPCR (B). mRNA levels in resting WT T cells were set to 1. (C and D) Glycolysis and glutaminolysis as determined by the generation of $^3\text{H}_2\text{O}$ from [^3H]-glucose (C) and the generation of $^{14}\text{CO}_2$ from [^{14}C]-glutamine (D), respectively.

Error bars in (B)–(D) represent standard deviation from the mean of triplicate samples. Data are representative of two independent experiments.

(E) Cell proliferation of resting (Rest), active WT, and KO T cells (48 hr) was determined as CFSE dilution.

(F) Cell size of resting (Rest), active WT, and KO T cells (24 hr) was determined as forward light scatter.

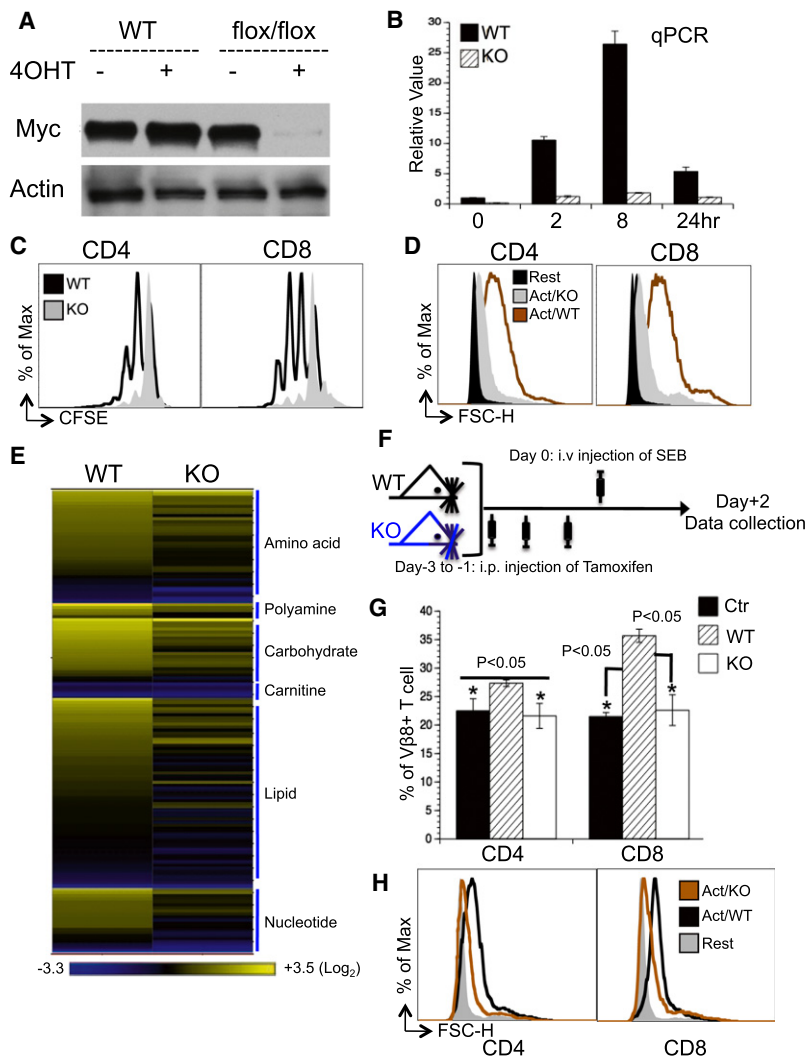
See also Figure S3.

Myc-Dependent Glutaminolysis Cross-talks with the mTOR Pathway

Deletion of Myc profoundly inhibited the upregulation of glutamine oxidation in active T cells (Figure 6A). Recent studies in human cell lines have suggested that enforced expression of Myc regulates the rate-limiting glutaminolytic enzyme glutaminase 1 (Gls1) through both transcription and posttranscriptional mechanisms (Gao et al., 2009; Wise et al., 2008), and p53 regulates the transcription of glutaminase 2 (Gls2) (Hu et al., 2010; Suzuki et al., 2010). We observed that neither the mRNA nor protein levels of Glis1 were elevated upon T cell activation

(Figures 2B and S2B), whereas the transcription and translation of glutaminase 2 (Gls2) along with other metabolic genes in glutamine catabolic pathways were induced in a Myc-dependent manner in vitro and in vivo (Figures 5D, 6B, and S5F). However, we found that p53 was dispensable for the induction of Glis2 and glutaminolysis upon T cell activation (Figures 6D and 6E).

Either glutamine starvation or Myc deletion was sufficient to abrogate activation-induced T cell growth (Figures S1D, S4D, and S4H). Because glutamine flux might regulate cell growth through the mTOR pathway (Nicklin et al., 2009),



glutamine-dependent modulation of mTOR activation might partially account for the regulation of T cell growth by Myc. Whereas the phosphorylation kinetics of upstream components (mTOR and AMPK) were unaffected, acute deletion of Myc impaired the activation of downstream effectors in the mTOR pathway, as indicated by the reduction of phosphorylation of P70K and the reduction of phosphorylation and protein levels of 4E-BP and S6 in Myc-deficient cells (Figure S6A). Consistent with an indispensable role for the glutamine antiporter CD98, a heterodimer of Slc3a2/Slc7a5, in modulating the mTOR pathway (Nicklin et al., 2009), both the transcription of Slc3a2/Slc7a5 and the cell surface appearance of CD98 were impaired in Myc-deficient T cells upon activation in vitro and in vivo (Figures 6B, 6C, S6C, and 5D). Moreover, glutamine starvation largely recapitulated the impairment of the activation of downstream effectors in the mTOR pathway as observed in Myc-deficient cells (Figure S6B). These results demonstrate an essential role for Myc in the upregulation of glutaminolysis upon activation and imply that cross-talk with the mTOR pathway via amino acid uptake may be involved in the Myc-dependent regulation of T cell growth.

Figure 4. Myc Is Required for T Cell Growth and Proliferation

(A) T cells isolated from either RosaCreERTam-Myc^{WT/WT} mice (WT) or RosaCreERTam-Myc^{flox/flox} mice (flox/flox) were pretreated with vehicle (–4OHT) or with 500 nM 4OHT (+4OHT). Myc protein levels were determined by immunoblot.

(B) Myc mRNA levels in T cells collected at the indicated times after activation as determined by qPCR. mRNA levels in resting WT T cells were set to 1. Error bars represent standard deviation from the mean of triplicate samples. Data are representative of two independent experiments.

(C) Cell proliferation of active T cells (48 hr) was determined as CFSE dilution.

(D) Cell size of resting (Rest) and active T cells (24 hr) was determined as forward light scatter.

(E) The level of intracellular metabolites in the indicated samples; the value of each metabolite was obtained from the average of triplicates and the level of each metabolite in resting WT T cells was set to 1. The heat map represents the log₂ value of the relative level of each metabolite, which was grouped in the indicated metabolic pathway (see color scale). The complete metabolomic profile is provided in Table S6.

(F–H) The experimental procedure (F) used for the SEB superantigen challenge model. Myc^{flox/flox} mice (WT) or RosaCreERTam-Myc^{flox/flox} mice (KO) were treated with tamoxifen (i.p. 1 mg/mouse for 3 days), and then were injected with SEB (i.v. 100 μg/mouse). The percentage (G) and the cell size (H) of Vβ8⁺ T cells were determined 2 days after injection of SEB.

See also Figure S4.

Myc-Driven Glutamine Catabolism Couples with Multiple Biosynthetic Pathways

Myc drives the expression of Gfpt1, Cad, Ppat, and Oat (Figures 2B and 6B), which not only are required for glutaminolysis but also control

critical steps in the hexosamine, pyrimidine, purine, and polyamine biosynthetic pathways (Figure 2B, left). Although we failed to detect any intermediate metabolites in the hexosamine biosynthetic pathway because of technical limitations, the levels of intermediate metabolites in the polyamine and nucleotide synthetic pathways were lower in Myc-deficient cells (Figures S6D and S6E; Table S6). Consistent with a previous study (Bowlin et al., 1987), difluoromethylornithine (DFMO), a potent inhibitor of ornithine decarboxylase (ODC) and polyamine biosynthesis, inhibited activation-induced T cell proliferation in vitro and in vivo (Figure 6F and 1H), and the addition of exogenous polyamines completely restored proliferation in the presence of DFMO in vitro (Figure 6F).

Addition of either hypoxanthine and thymidine (HT) or polyamines (PAs) alone was insufficient to permit cell proliferation in the absence of glutamine (Figure S7A). Because α-KG is required to maintain cell viability and minimal cell proliferation in the absence of glutamine (Wise et al., 2008), addition of α-KG alone was sufficient to drive only one cell division in the absence of glutamine. However, the combination of α-KG with either HT or with PAs promoted additional cell divisions and

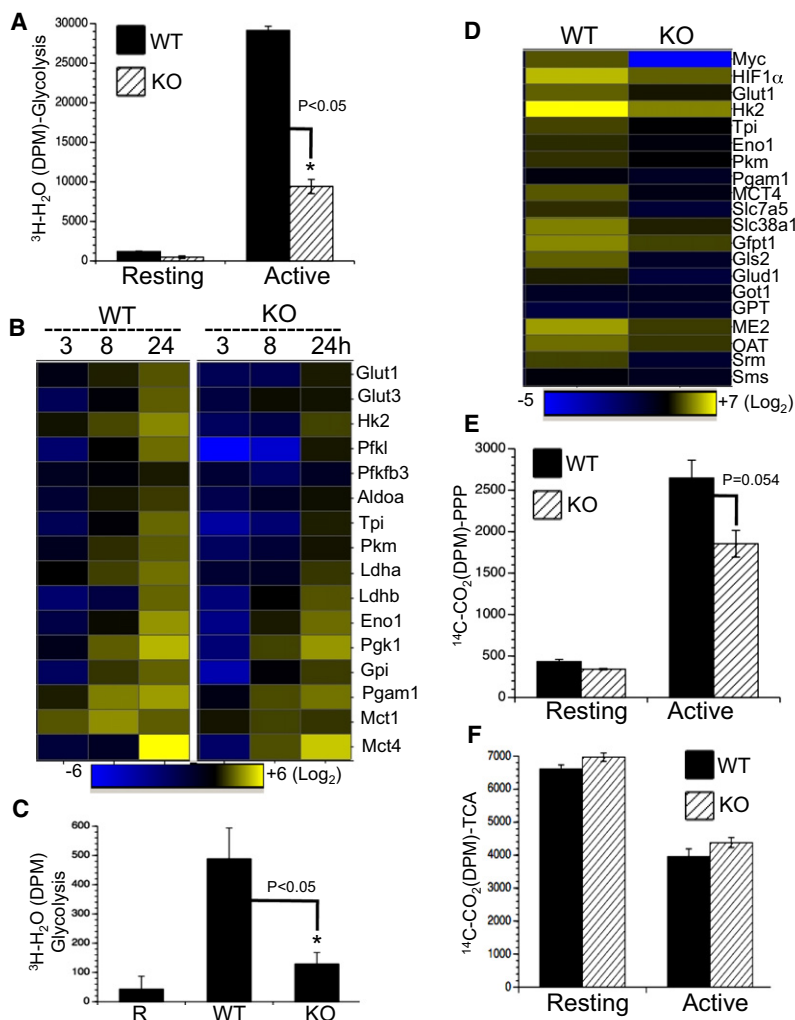


Figure 5. Myc Drives a Transcription Program that Regulates Glucose Catabolism upon T Cell Activation

(A) Glycolytic flux as determined by the generation of $^3\text{H}_2\text{O}$ from $[3\text{-}^3\text{H}]\text{-glucose}$.

(B) qPCR analyses of metabolic genes. mRNA levels in resting WT T cells were set to 1. The heat map represents the $\log_2$ value of the relative mRNA expression level (see color scale). Values and standard deviations are provided in Table S5. Data are representative of two independent experiments, performed in triplicate.

(C and D) Naive $\text{CD4}^+\text{V}\beta 8^+$ T cells (R) were sorted from mice without SEB immunization; WT and acutely Myc-deleted $\text{CD4}^+\text{V}\beta 8^+$ T cells (KO) were isolated from mice 2 days after SEB immunization (Figure 4F). Glycolysis (C) and mRNA expression (D and Table S5) in the indicated groups were determined by the generation of $^3\text{H}_2\text{O}$ from $[3\text{-}^3\text{H}]\text{-glucose}$ and qPCR, respectively.

(E and F) PPP flux (E) and pyruvate oxidation through the TCA cycle (F) were determined by the generation of $^{14}\text{C}\text{-CO}_2$ from $[1\text{-}^{14}\text{C}]\text{-glucose}$ and from $[2\text{-}^{14}\text{C}]\text{-pyruvate}$, respectively.

Error bars in (A), (E), and (F) represent standard deviation from the mean of triplicate samples. See also Figure S5.

growth (Figures 6G and S7B). Consistent with a direct role of Myc in regulating cell proliferation (Dang et al., 2006; Eilers and Eisenman, 2008), addition of $\alpha\text{-KG}$ with HT and PAs failed to promote cell proliferation in Myc-deficient T cells (Figure S7C). Taken together, our results indicate that Myc-driven glutaminolysis coordinates multiple biosynthetic pathways to support activation-induced cell proliferation and growth.

A Myc-Dependent Metabolic Pathway Linking Glutaminolysis to Ornithine and Polyamine Biosynthesis

Although ornithine is produced largely from arginine via the action of arginase as part of the urea cycle (Morris, 2004), urea cycle enzymes including arginase were not induced upon activation (Figure S7D) nor were any defects in proliferation observed in T cells lacking arginase I (Figures 7A and 7B), the dominant isoform in the thymus (Yu et al., 2003). These results indicate that arginine is not the only precursor for ornithine in T cells. Previous studies in bovine endothelial and intestinal epithelial cells suggested the possible presence of a noncanonical metabolic pathway, which generates ornithine from glutamine (Wu et al., 2000; Wu and Morris, 1998). This putative pathway is composed of two key steps: aldehyde dehydrogenase 18 family

member A1 (Aldh18a1) converts glutamate to (S)-1-pyrroline-5-carboxylate (P-5-C), after which OAT converts P-5-C to ornithine (Figure 7C). Alternatively, proline has also been suggested as a source for ornithine through a two-step reaction involving proline dehydrogenase (Prodh) and OAT (Phang et al., 2008). Intriguingly, the metabolic enzymes Aldh18A1, Prodh, and OAT that link glutamine and proline to ornithine synthesis were induced in a Myc-dependent manner upon T cell activation (Figures 7C and S5F), as were the metabolic enzymes, ODC, spermine synthase (SRM), and

spermidine synthase (SMS), that convert ornithine to polyamines (Figure 7C). Therefore, we followed $\text{U-}^{13}\text{C}$ -isotope labeled glutamine ($\text{U-}^{13}\text{C}$ -glutamine) flux and observed the incorporation of ^{13}C into all five carbons of ornithine ($\text{U-}^{13}\text{C}$ -ornithine), which accounted for up to 30% of total ornithine in active T cells but only 1% in resting T cells (Figure S7E). Finally, deletion of Myc partially abolished the incorporation of glutamine-derived carbons into both ornithine and putrescine (Figure 7D). Taken together, our results indicate a Myc-dependent noncanonical ornithine biosynthetic pathway coupling glutaminolysis to ornithine and polyamine biosynthesis in activated T cells (Figure 7E).

DISCUSSION

Extracellular signals play an essential role in dictating eukaryotic cell growth, proliferation, and metabolism. TCR- and costimulator-mediated signals drive T cell growth and clonal expansion, and coordinately reprogram metabolic cascades to fulfill the ensuing bioenergetic and biosynthetic demands (Fox et al., 2005; Krauss et al., 2001). Although ectopic overexpression of Myc in transformed cell lines coordinately drives cell proliferation and proliferation-associated metabolic activities, including the

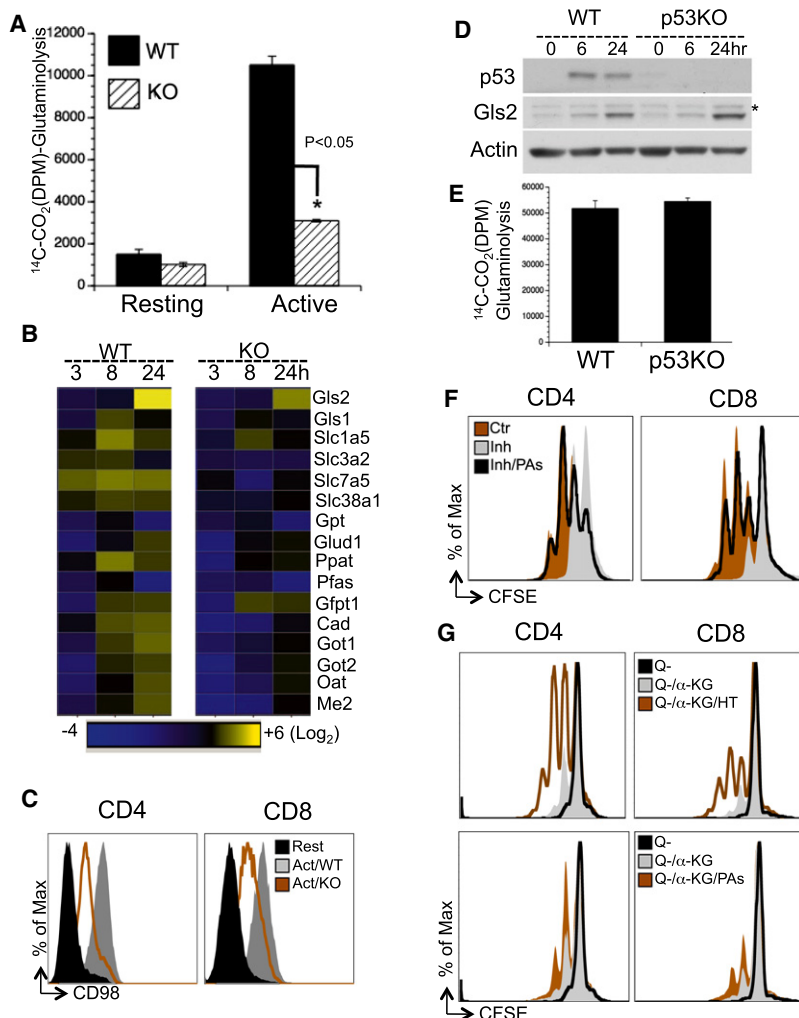


Figure 6. Myc Drives a Transcriptional Program Coupling Glutaminolysis to Biosynthetic Pathways

(A) Rates of glutaminolysis, determined by the generation of $^{14}\text{C-CO}_2$ from $[\text{U-}^{14}\text{C}]$ -glutamine. Data are representative of two independent experiments.

(B) qPCR analyses of metabolic genes. mRNA levels in resting WT T cells were set to 1. The heat map represents the $\log_2$ value of the relative mRNA expression level (see color scale). Values and standard deviations are provided in Table S5. Data are representative of two independent experiments.

(C) Cell surface expression of CD98 was determined by flow cytometry.

(D) Protein levels of Glis2 and p53 were determined by immunoblot. The asterisk indicates a nonspecific band.

(E) Glutaminolytic flux was determined by the generation of $^{14}\text{C-CO}_2$ from $[\text{U-}^{14}\text{C}]$ -glutamine.

(F) Cell proliferation of active T cells (48 hr), treated with vehicle (Ctr), with 1 mM DFMO (Inh) or with 1 mM DFMO plus polyamine mixture (Inh/PAs), determined by CFSE dilution. The composition and concentration of polyamine mixture are provided in Table S7.

(G) T cells were collected at 72 hr after activation in glutamine-free media (Q-), in glutamine-free media supplemented with 2 μM α -KG (Q-/α-KG), in glutamine-free media supplemented with 2 μM α -KG and 100 μM hypoxanthine and 16 μM thymidine (Q-/α-KG/HT), or in glutamine-free media supplemented with 2 μM α -KG and polyamine mixture (Q-/α-KG/PAs). The composition and concentration of the polyamine mixture is provided in Table S7. Cell proliferation was determined as CFSE dilution by flow cytometry.

Error bars in (A) and (E) represent standard deviation from the mean of triplicate samples. See also Figure S6.

catabolism of glucose and glutamine (Dang and Semenza, 1999; Gordan et al., 2007), whether the induction of endogenous Myc upon physiological stimulation plays a similar role has not been explored. We show that the activation of T cells rapidly switches metabolic programs from fatty acid β -oxidation and pyruvate oxidation via the TCA cycle to aerobic glycolysis, PPP, and glutaminolysis. This metabolic reprogramming is associated with a global change in the metabolic transcriptome, which follows the induction of endogenous Myc and HIF1 α upon activation. Based on the effects of acute genetic ablation of these transcription factors, we identified Myc as a critical metabolic reprogramming factor in T cells. Our studies shed light on the complex utilization of metabolic resources, pathways, and intermediates that rationalize the strict dependence of T cell growth and proliferation on glutaminolysis and indicate a glutaminolysis- and Myc-dependent de novo polyamine biosynthetic pathway essential for T cell proliferation. Whereas previous studies clearly demonstrate that Myc directly regulates T cell proliferation mainly through transcriptional control of cell cycle regulators (Dose et al., 2006; Iritani et al., 2002), our studies implicate Myc as an essential coordinator of T cell activation-induced cell growth, proliferation, and several aspects of metabolic reprogramming.

T cell activation-induced glycolysis and glutaminolysis is reminiscent of the metabolic changes in tumor cells, where both aerobic glycolysis (the Warburg effect) and glutaminolysis are driven by aberrant oncogenic signals (Dang and Semenza, 1999; Vander Heiden et al., 2009). Notably, ectopic overexpression of Myc not only alters the transcription of several glycolytic genes but also preferentially switches the expression of PKM1 to PKM2 by modulating the alternative splicing of the PKM transcript (Dang and Semenza, 1999; David et al., 2010). Although the transcription of metabolic enzymes and transporters involved in almost every step of glycolysis including PKM2 are induced in a Myc-dependent manner upon T cell activation, whether active T cells preferentially express PKM2 through alternative splicing remains to be determined. In contrast, the metabolic program associated with glutaminolysis in tumor cell lines and activated T cells is slightly different. In tumor cells, Myc controls glutamine uptake through transcription of SLC3A2, SLC5A1, and SLC7A1 and regulates the conversion of glutamine to glutamate through both transcriptional and posttranslational regulation of GLS1 (Gao et al., 2009; Wise et al., 2008). Corresponding steps in active T cells, however, are controlled by Myc-mediated transcription of SLC32A1, SLC32A2, and GLS2. Although recent studies indicate that the tumor suppressor p53 is required for the induction of GLS2 in a human tumor cell line (Hu et al., 2010; Suzuki et al., 2010),

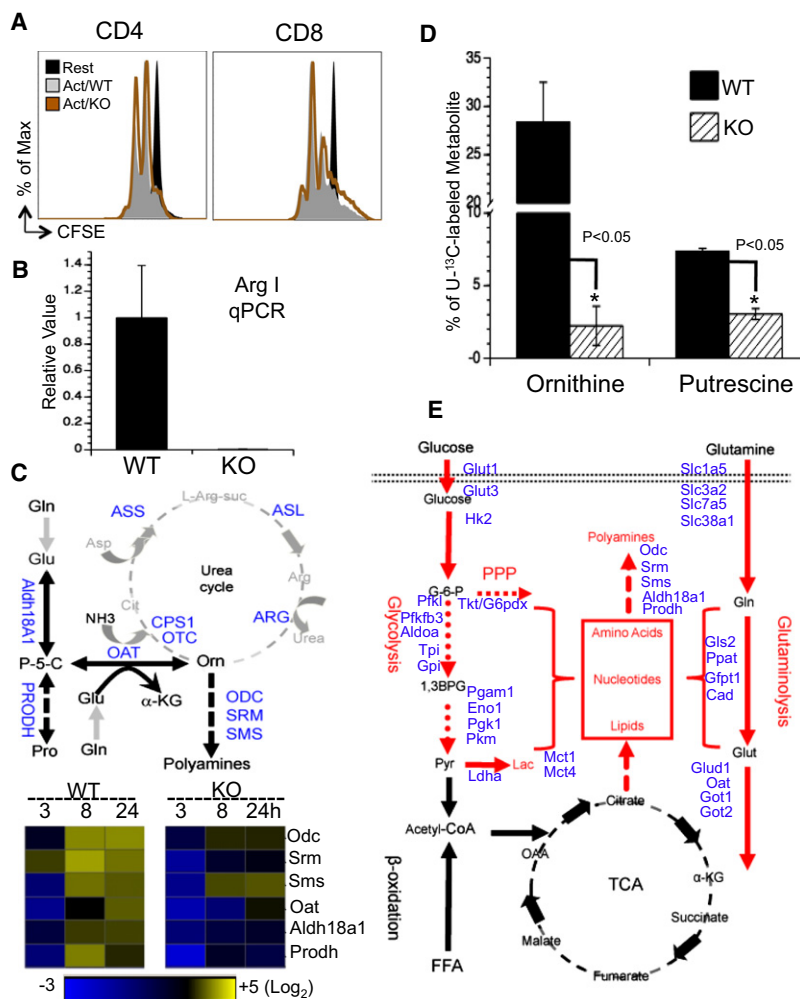


Figure 7. Myc-Driven Glutaminolysis Fuels Polyamine Biosynthesis upon T Cell Activation

(A) Arginase I WT and KO T cells were collected 48 hr after activation (Act/WT and Act/KO, respectively). Cell proliferation was determined by CFSE dilution.

(B) WT and arginase I KO active T cells were analyzed by qPCR for arginase I mRNA expression.

(C) Top: Potential metabolic steps linked to the polyamine biosynthetic pathway, with metabolic genes measured highlighted in blue. Bottom: WT and Myc KO T cells were collected at the indicated times after activation and qPCR analyses of metabolic genes was performed. mRNA levels in resting T cells were set to 1. The heat map represents the log₂ value of the relative mRNA expression level (see color scale). Values and standard deviations are provided in Table S5. Data are representative of two independent experiments, performed in triplicate.

(D) T cells were cultured in media containing U-¹³C-glutamine and collected at the indicated times after activation. The intracellular levels of ornithine and putrescine including U-¹³C- and ¹²C-labeled forms were determined by mass spectroscopy.

(E) Myc-dependent metabolic reprogramming upon T cell activation, with upregulated metabolic pathways highlighted in red, downregulated metabolic pathways highlighted in black, and Myc-dependent metabolic genes highlighted in blue.

Error bars in (B) and (D) represent standard deviation from the mean of triplicate samples. See also Figure S7.

we have found that the upregulation of GLS2 in murine T cells is p53 independent. Although the regulatory mechanisms involved are slightly different, the resemblance of tumor-associated and T cell activation-associated metabolic programs indicate that a general metabolic switch is required to support cell growth and proliferation in both pathological and physiological situations.

Glutamine has long been known to be an indispensable nutrient for proliferating cells (Kovacevic and McGivan, 1983). Enhanced glutamine utilization in rat lymphocytes upon polyclonal mitogen stimulation is responsible for providing precursors for nucleotide biosynthesis (Newsholme et al., 1985a). Metabolic flux analysis in a glioblastoma cell line further revealed that a major portion of glutamine-derived nitrogen and carbon are incorporated into amino acids and lactate, a fraction of which are secreted as metabolic by-products (DeBerardinis et al., 2007). These findings indicate that glutamine is an important nitrogen and carbon donor for a variety of biosynthetic precursors. Supporting this notion, we found that activation of T cells triggers a Myc-dependent induction of glutaminolysis, which not only leads to the production of α-KG in the TCA cycle but is also coupled to nucleotide biosynthesis. Furthermore, the depletion of intracellular ornithine and polyamines in Myc-deficient activated T cells impli-

cates a noncanonical, Myc-dependent polyamine biosynthetic pathway upon activation.

Early studies showed that overexpression of Myc in transformed cell lines induces the expression of ODC (Bello-Fernandez, et al., 1993). Although the protein level of ODC was not elevated, the mRNAs of all three metabolic enzymes downstream of ornithine were induced

in a Myc-dependent manner upon T cell activation, leading to a massive increase in polyamine synthesis. The Myc-dependent induction of Aldh18a1, Prodh, and OAT further indicates a Myc-dependent metabolic route upstream of ornithine that links glutamine and possibly proline to ornithine. Consistent with this, the Myc-dependent incorporation of ¹³C-labeled glutamine carbon into ornithine after T cell activation indicates that activated T cells may acquire an alternative Myc-dependent pathway coupling glutaminolysis to polyamine biosynthesis to meet the markedly increased polyamine demands required for proliferation. Our results suggest that Myc-dependent glutamine catabolism not only replenishes the intermediate metabolites of the TCA cycle by producing the anaplerotic substrate α-KG, but also coordinates with Myc-dependent glucose catabolism to support amino acid, nucleotide, and lipid biosynthesis. In addition, glutamine catabolism is tightly coupled to the biosynthesis of polyamines that are essential for T cell proliferation.

EXPERIMENTAL PROCEDURES

Animals

Hif1a^{flox/flox} and p53-null mice on the C57BL/6 background were purchased from the Jackson Laboratory (Jacks et al., 1994; Ryan et al., 2000) and *Hif1a*^{flox/flox} mice were crossed with ROSA26CreERT2 (de Luca et al., 2005).

Myc^{flox/flox} mice were obtained from F. Alt (Harvard) (de Alboran et al., 2001) and were backcrossed five generations onto the C57BL/6 background before breeding to ROSA26CreER(Tam) (Badea et al., 2003). QA Resulting offspring were interbred to obtain experimental animals. *Arg^{flox/flox}*; Tie2Cre mice that were backcrossed to the C57BL/6 background were obtained from P. Murray (SJCRRH, Memphis) (El Kasmi et al., 2008). Mice at 7–12 weeks of age were used. All mice were kept in specific-pathogen-free conditions within the Animal Resource Center at St. Jude Children's Research Hospital. Animal protocols were approved by the Institutional Animal Care and Use Committee of St. Jude Children's Research Hospital.

RNA Isolation, Reverse Transcription, and qPCR

Total RNA was isolated with the RNeasy Mini Kit (QIAGEN) and was reverse transcribed with random hexamer primers and M-MLV Reverse Transcriptase (Invitrogen). Real-time qPCR for cytokines was performed with a TaqMan Low Density Array (4367786, ABI). The comparison of cytokine gene expression in resting and active T cells was visualized as heat maps with Cluster (Eisen and Brown, 1999) and TreeView (Eisen et al., 1998). SYBR Green-based quantitative RT-PCR for other genes was performed with the Applied Biosystems 7900 Real Time PCR System. Samples for each experimental condition were run in triplicate and were normalized to β -2-microglobulin to determine relative expression levels. Primer sequences were obtained from PrimerBank (Spandidos et al., 2010) except for Gls1 and Gls2, which were manually designed. Primer sequences are listed in Table S7. Comparison of the expression of selected metabolic genes in resting and active T cells was visualized as heat maps with Spotfire.

Promoter Analyses

Transcription factor motifs enriched in the promoters were identified by using the motif analysis algorithm in the PASTAA webserver (<http://trap.molgen.mpg.de/cgi-bin/pastaa.cgi>). This method considers the input order of promoters and uses a biophysical model of promoter-transcription factor interaction to detect motifs (Roeder et al., 2009). The input order was that of decreasing fold-change in expression and default settings of the server were used.

Metabolic Assays

Glycolytic flux was determined by measuring the detritiation of [3-³H]-glucose. Fatty acid β -oxidation flux was determined by measuring the detritiation of [9,10-³H]-palmitic acid. Glutamine oxidation flux was determined by the rate of ¹⁴CO₂ released from [U-¹⁴C]-glutamine. Pyruvate oxidation flux was determined by the rate of ¹⁴CO₂ released from [2-¹⁴C]-pyruvate. Glucose oxidation flux through the pentose phosphate pathway (PPP) was determined as previously described with some modifications described in the Supplemental Information. Respiration was measured in intact T cells with the Seahorse XF24 analyzer. Metabolomic profiling was performed by Metabolon (Durham, NC). Metabolic flux analysis of ¹³C-glutamine in active T cells was determined by LC-MS.

Statistical Analysis

p values were calculated with Student's t test. p values < 0.05 were considered significant.

SUPPLEMENTAL INFORMATION

Supplemental Information includes Supplemental Experimental Procedures, seven figures, and seven tables and can be found with this article online at doi:10.1016/j.immuni.2011.09.021.

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